

Project Proposal: PocketBirdNET—Distilling a State-of-the-Art Bird Classifier onto a Microcontroller

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1 Problem Statement

State-of-the-art bird sound classifiers such as BirdNET (~50 MB, ~6,000 species) require a phone or a cloud connection to run. Field biologists, conservation stations, and remote acoustic monitoring posts often need fully offline operation: no internet, no phone tethering, and milliwatt-scale power budgets for long deployments.

The TinyML question this project addresses is whether a state-of-the-art audio classifier can be compressed by roughly three orders of magnitude—from ~50 MB down to under 100 KB—to fit on the Arduino Nano 33 BLE Sense (256 KB RAM, 1 MB flash, Cortex-M4) while preserving usable accuracy on a focused species set. The contribution is a systematic comparison of three training strategies for the on-device student model: from-scratch training, transfer learning from MobileNet, and knowledge distillation from BirdNET.

2 Dataset

Source: Xeno-Canto, an open archive of bird vocalizations: <https://xeno-canto.org/>

Contents: The dataset will be assembled by querying Xeno-Canto for ten Pacific Northwest species plus a background/no-bird class drawn from environmental noise recordings:

- **Target species (10):** American Robin, Black-capped Chickadee, Steller’s Jay, Northern Flicker, Song Sparrow, Anna’s Hummingbird, Dark-eyed Junco, American Crow, Pacific Wren, House Finch.
- **Background class (1):** ambient outdoor recordings without target species, including wind, traffic, and human voice segments.
- **Volume target:** 200–500 clips per class after filtering by recording quality.

Preprocessing:

- Resample to 16 kHz mono; segment into 3-second windows with 50% overlap.
- Compute mel-spectrograms (40 mel bins, ~32 frames per window) as model input.
- Augmentation: time-shift, gaussian noise injection, pitch perturbation, mixup with background recordings.
- Splits at the recording level (not window level) to prevent temporal leakage: 70% train, 15% validation, 15% test, stratified by species.

Limitations:

- *Recording-quality variance:* Xeno-Canto contributions vary in microphone, distance, and background noise.
- *Domain mismatch:* Xeno-Canto recordings are typically clean and close-mic’d, while the Nano’s PDM microphone in real conditions will be noisier.
- *Class imbalance:* Some species are over-represented; mitigated via stratified sampling and

per-class clip caps.

3 Modeling Approach

3.1 Input Modality and Features

Audio captured via the Nano 33 BLE Sense onboard PDM microphone. On-device preprocessing computes mel-spectrograms using the audio frontend from the TensorFlow Lite Micro keyword-spotting examples, adapted to a 3-second window.

3.2 Student Architecture

- 2D CNN over mel-spectrogram input ($40 \times 32 \times 1$).
- 4-5 depthwise-separable convolutional blocks following the MobileNet design (depthwise + pointwise convolution per Lecture 4).
- Width multiplier $\alpha < 1$ to fit the 256 KB RAM ceiling.
- Global average pooling $\rightarrow$ small fully-connected layer $\rightarrow$ 11-class softmax.
- Target footprint: under 100 KB after INT8 quantization.

3.3 Training Variants and Compression

The core contribution is a head-to-head comparison of four model variants on identical data and architecture:

- **A. Scratch:** Student CNN trained from random initialization on hard labels only.
- **B. Transfer learning:** MobileNet-pretrained backbone, frozen early layers, fine-tuned final layers (per Lecture 4 transfer learning workflow).
- **C. Knowledge distillation:** Hinton-style soft-label distillation from BirdNET (<https://github.com/kahst/BirdNET-Analyzer>) as teacher; teacher logits filtered to the 11 target classes and re-normalized; loss is a weighted combination of distillation loss (temperature $T = 4$) and hard-label cross-entropy.
- **D. Distilled + pruned + quantized:** The deployment variant. Variant C, then magnitude pruning at 30-50% sparsity using the TensorFlow Model Optimization Toolkit, then INT8 post-training quantization via the TFLite converter. Quantization-aware training (QAT) used as a fallback if PTQ accuracy drops below acceptable threshold.

4 Deployment Plan

4.1 Hardware and Pipeline

- **Board:** Arduino Nano 33 BLE Sense (provided in class kit).
- **Display:** SSD1306 OLED over I²C for live species name and confidence.
- **Software pipeline:** TensorFlow (training) $\rightarrow$ TFLite (conversion) $\rightarrow$ TFLite Micro (Nano deployment), matching the workflow from Lecture 4.
- **On-device flow:** PDM mic $\rightarrow$ rolling audio buffer $\rightarrow$ mel-spectrogram $\rightarrow$ CNN inference $\rightarrow$ softmax $\rightarrow$ confidence threshold ($\geq 70\%$) $\rightarrow$ OLED display; below-threshold predictions render as “Unknown.”

4.2 Evaluation

- **Held-out test accuracy:** top-1 accuracy, macro F1, per-class F1, and confusion matrix on the 15% test split. Target: >80% top-1, >0.75 macro F1.
- **Compression frontier:** a single results table reporting accuracy, model size (KB), peak RAM (KB), and inference latency (ms) for all four variants (A–D), measured on the Nano.
- **Live demo:** bird call recordings played near the Nano during the presentation; OLED shows correct classification in real time. Background sounds (typing, voices, music) used to verify the “Unknown” fallback.
- **Deliverables:** GitHub repository with training notebooks, TFLite models, Arduino sketch, results, and demo video.

5 Challenges and Solutions

- **256 KB RAM ceiling:** mitigated via depthwise-separable convolutions, width multiplier $\alpha < 1$, INT8 quantization, and pre-deployment verification of peak tensor sizes using the TFLite Micro memory planner.
- **On-device mel-spectrogram in C:** reuse and adapt the existing audio frontend from the Arduino_TensorFlowLite keyword-spotting examples rather than implementing from scratch.
- **BirdNET label remapping for distillation:** filter BirdNET’s $\sim 6,000$ -class output to our 11 target classes and re-normalize via softmax before computing distillation loss; verify alignment on a small audited subset.
- **Domain mismatch (clean Xeno-Canto vs. noisy Nano microphone):** heavy augmentation during training, confidence threshold for “Unknown,” and a small self-collected validation set of ambient recordings to estimate real-world robustness.
- **Distillation may not outperform baselines:** the four-way comparison stands on its own as a contribution; a null or negative distillation result is still a valid and reportable finding.
- **Past-class overlap:** a Spring 2024 EEP596 project (“What the Chirp?”) addressed bird classification on similar hardware. Our project is methodologically distinct: where prior work used a single Edge Impulse pipeline, we systematically compare three training strategies and report a full compression frontier.

6 Team Plan

- **Rishabh Goenka:** ML pipeline—dataset curation from Xeno-Canto, BirdNET teacher logit extraction, student architecture design, all four training runs.
- **Aarav Wadhvani:** Compression and embedded deployment—TFLite conversion, INT8 quantization, magnitude pruning, deployment to the Nano, on-device audio frontend integration.
- **Carson Large:** Evaluation and demo—benchmarking scripts, results tables, OLED interface and confidence-threshold logic, live demo preparation, proposal and final report writing.